

- Order matters

Operators do not commute in general, so that e.g.

$$\langle \psi | \hat{A} \hat{B} | \psi \rangle \neq \langle \psi | \hat{B} \hat{A} | \psi \rangle$$

This can be easily seen in the position rep. for $\hat{x}$ & $\hat{p}_x$ ~~is~~

$$\langle x | \hat{x} \hat{p}_x | \psi \rangle = -i\hbar x \frac{\partial}{\partial x} \psi(x)$$

$$\langle x | \hat{p}_x \hat{x} | \psi \rangle = -i\hbar \frac{\partial}{\partial x} (x \psi(x)) = -i\hbar x \frac{\partial}{\partial x} \psi(x) - i\hbar \psi(x)$$

$$\Leftrightarrow \langle x | [\hat{x}, \hat{p}_x] | \psi \rangle \equiv \langle x | \hat{x} \hat{p}_x - \hat{p}_x \hat{x} | \psi \rangle = \langle x | i\hbar | \psi \rangle$$

or $\boxed{[\hat{x}, \hat{p}_x] = i\hbar}$ ^{op.} very important!

Notice that, e.g.

$$[\hat{y}, \hat{p}_x] = 0$$

§ 5. Quantum physics.

Let's ~~exp~~ define the Hamiltonian operator.

$$\hat{H} \equiv \frac{\hat{p}^2}{2m} + \hat{V}$$

and write the stationary Schrödinger equation as

$\hat{H}|\psi\rangle = E|\psi\rangle$ ^{eliminate} $\Rightarrow \boxed{\nabla^2 \psi(r) + \frac{2m}{\hbar^2} (E - V) \psi(r) = 0}$

~~Many~~ of the typical physical systems can be accommodated to be described by this equation.

It's surprising the quantum physics that it reveals. We shall see a couple of toy examples before coming to the description of real systems.

~ Particle confined

Suppose a 1D system in which a particle is free to move only within the interval $0 \leq x \leq L$.

How can we describe this situation?

$$\rightarrow V(x) = \begin{cases} 0 & 0 \leq x \leq L \\ \infty & \text{otherwise} \end{cases}$$

This potential avoids a particle to escape from the allowed region. However, we must impose an additional condition that warrants that

$$\psi(x) = 0 \quad x \leq 0, x \geq L$$

Namely:

$$\psi(x) = 0 \quad x \leq 0, x \geq L \quad \text{Boundary cond.}$$

The system can be solved now. Within the interval, we have

$$\frac{d^2\psi}{dx^2} + \underbrace{\frac{2m}{\hbar^2} E}_{\equiv k^2} \psi = 0 \quad \Leftrightarrow \quad \frac{d^2\psi}{dx^2} + k^2 \psi = 0$$

oscillator equation

$$\Rightarrow \psi(x) = A \sin kx + B \cos kx$$

Demanding $\psi(x) = 0$ @ $x = 0$:

$$\psi(x=0) = A \sin kx \Big|_{x=0} + B \cos kx \Big|_{x=0} = \underline{\underline{B = 0}}$$

Demanding $\psi(x) = 0$ @ $x = L$:

$$\psi(x=L) = A \sin kL = 0$$

which can be satisfied either if $A = 0 \rightarrow$ trivial sol. (no particle)

or if

$$kL = n\pi \quad \Leftrightarrow \quad \frac{\sqrt{2mE}}{\hbar} L = n\pi \quad \Leftrightarrow \quad \boxed{E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}} \quad n \in \mathbb{N}^+$$

$$\& \quad \psi_n(x) = A \sin \frac{n\pi x}{L}$$

The normalization factor can be found using

$$1 = \int_{-\infty}^{\infty} P_n dx = \int_{-\infty}^{\infty} dx |\psi_n|^2 = A^2 \int_0^L dx \sin^2\left(\frac{n\pi x}{L}\right) = A^2 \int_0^L dx \left(\frac{1}{2} - \frac{1}{2} \cos\left(\frac{2n\pi x}{L}\right)\right)$$

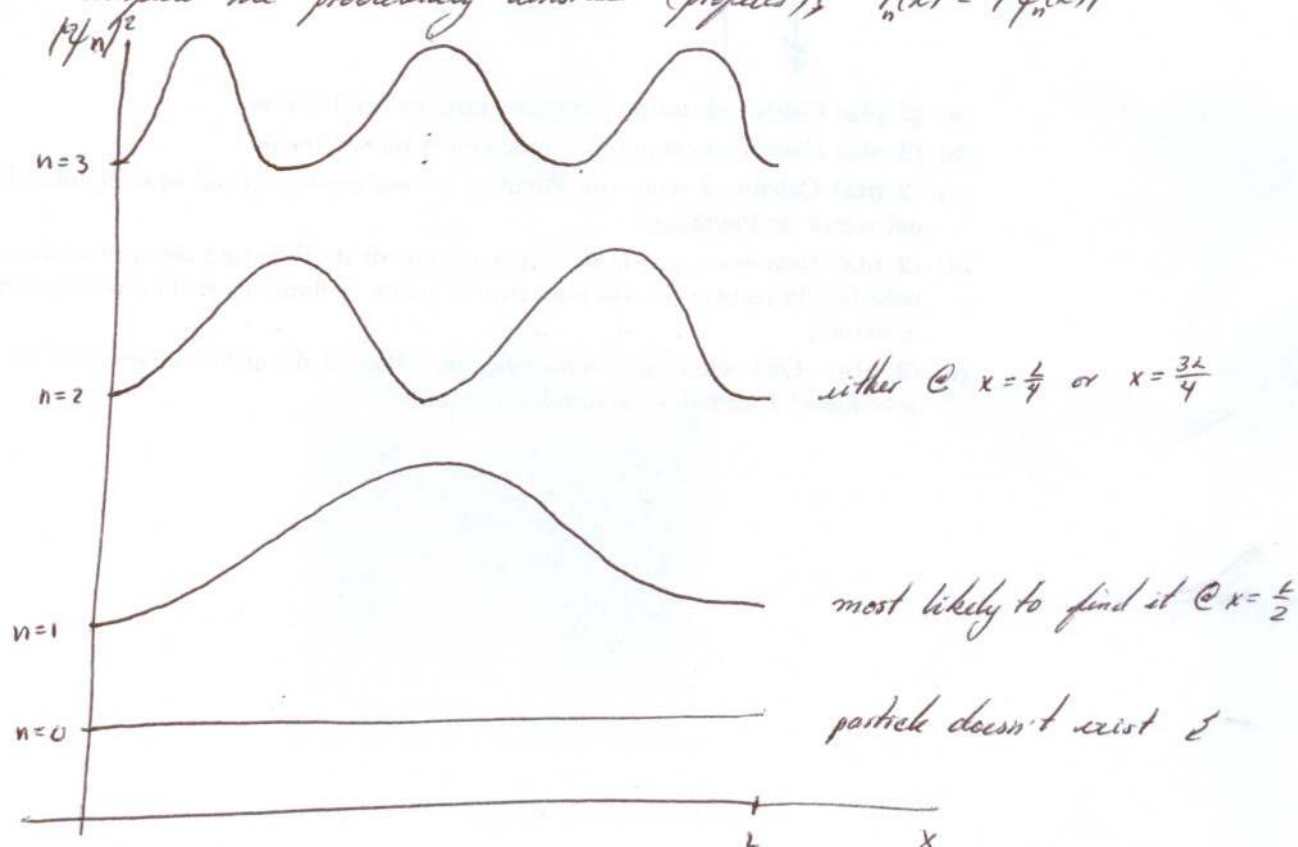
$$= \frac{A^2}{2} L - \frac{A^2}{2} \sin\left(\frac{2n\pi x}{L}\right) \frac{L}{2n\pi} \Big|_0^L = \frac{A^2 L}{2}$$

$$\Leftrightarrow A = \sqrt{\frac{2}{L}}$$

$$\Rightarrow \boxed{\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}}$$

This wavefunction is interpreted as the eigenfunction of the system with energy $E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$

One can compute the probability densities (profiles), $P_n(x) = |\psi_n(x)|^2$



which contradicts clearly classical physics, which would predict an equal probability to find the particle anywhere between $[0, L]$.

(36)

the average place
What's the most likely place to find the particle in quantum mechanics?

$$\begin{aligned}\langle \hat{x} \rangle &= \int_{-\infty}^{\infty} dx \psi_n^*(x) x \psi_n(x) = \int_0^L dx x |\psi_n(x)|^2 = \frac{2}{L} \int_0^L dx x \sin^2 \frac{n\pi x}{L} \\ &= \frac{2}{L} \int_0^L dx x \frac{1}{2} \left(1 - \cos \frac{2n\pi x}{L} \right) \\ &= \frac{2}{L} \left(\frac{L^2}{4} \right) = \frac{L}{2}\end{aligned}$$

What's the average momentum?

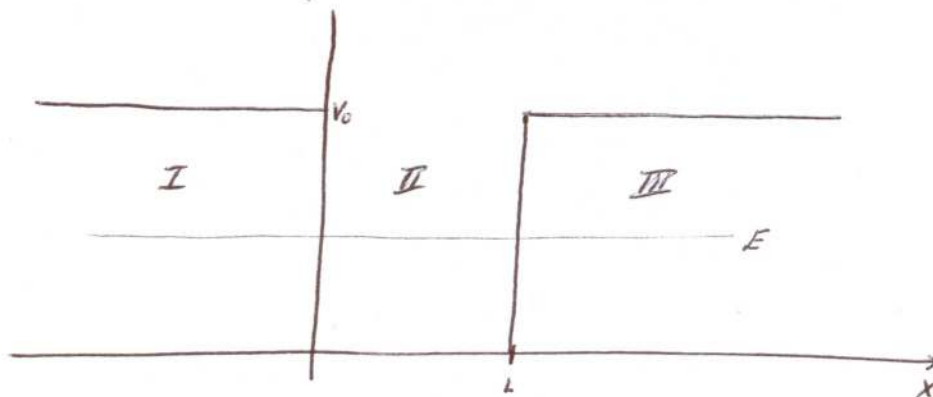
$$\begin{aligned}\langle \hat{p} \rangle &= \int_{-\infty}^{\infty} dx \psi_n^*(x) \left(-i\hbar \frac{d}{dx} \right) \psi_n(x) = -i\hbar \frac{2}{L} \int_0^L dx \sin\left(\frac{n\pi x}{L}\right) \frac{n\pi}{L} \cos \frac{n\pi x}{L} \\ &= -\frac{i\hbar}{L} \left(\sin^2 \frac{n\pi x}{L} \right) \Big|_0^L = 0\end{aligned}$$

using $\int dx \sin ax \cos bx = \frac{1}{2a} \sin^2 ax$. Thus, in average, the momentum of a particle is $\langle \hat{p} \rangle = 0$

~ Particle confined by a "realistic" barrier

A more realistic barrier ~~is~~ not infinitely hard, so that, if enough energy is applied to the particle, it can escape. This situation is depicted by a potential of the following form

$$V(x) = \begin{cases} 0 & 0 \leq x \leq L \\ V_0 & x \leq 0, x \geq L \end{cases}$$



$$V_0 - E > 0$$

We know that if $E > V_0$, the particle must behave as a free particle everywhere. The question is how it behaves when $V_0 - E > 0$.

As shown in the figure, we shall split this problem in three regions. In region II, we know the solution from the previous case

$$\psi_{II}(x) = A \sin(kx) + B \cos(kx) \quad k^2 = \frac{2mE}{\hbar^2}$$

whereas in regions I, III, we must solve the Schrödinger eq.

$$\frac{d^2 \psi}{dx^2} - \underbrace{\frac{2m}{\hbar^2} (V_0 - E)}_{\equiv \omega^2} \psi = 0$$

$$\omega^2 = \frac{2m}{\hbar^2} (V_0 - E) = \frac{2mV_0}{\hbar^2} - k^2 \equiv \beta^2 - k^2$$

whose solutions are

$$\psi_I = C e^{\omega x} + D e^{-\omega x}$$

$$\psi_{III} = E e^{\omega x} + F e^{-\omega x}$$

In regions I, III, we must impose

$$\psi_I(x \rightarrow -\infty) = 0 \quad \& \quad \psi_{III}(x \rightarrow \infty) = 0$$

$$\Rightarrow D = E = 0$$

$$\Rightarrow \begin{aligned} \psi_I &= C e^{\omega x} \\ \psi_{III} &= F e^{-\omega x} \end{aligned}$$

Further, we must impose the conditions

$$\psi_I(x=0) = \psi_{II}(x=0), \quad \psi_{II}(x=L) = \psi_{III}(x=L)$$

$$\& \quad \left. \frac{d\psi_I}{dx} \right|_{x=0} = \left. \frac{d\psi_{II}}{dx} \right|_{x=0}, \quad \left. \frac{d\psi_{II}}{dx} \right|_{x=L} = \left. \frac{d\psi_{III}}{dx} \right|_{x=L}$$

$\Leftrightarrow$

$$C = B$$

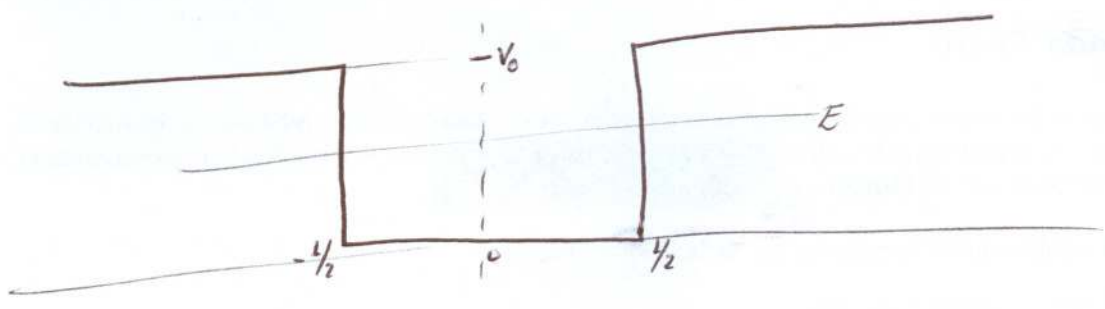
$$C\omega = Ak$$

$$A \sin kL + B \cos kL = F e^{-\omega L}$$

$$Ak \cos kL - Bk \sin kL = -F \omega e^{-\omega L}$$

$\updownarrow$

In order to be able to use symmetry arguments, it's better to state our potential



We then suppose
In the region $[-\frac{L}{2}, \frac{L}{2}]$ we have both even ($\cos kx$) and odd ($\sin kx$) solutions.
It's convenient to split the solutions then in even & odd due to the symmetry of the problem (Theorem: if $V(x) = V(-x)$, then $\psi(x)$ is even or odd.)

Even solutions

$$\psi(x) = \psi(-x)$$
$$\Rightarrow \psi_{II}(x) = B \cos(kx) \quad \& \quad C = F \quad \left(\begin{array}{l} \psi_I = C e^{\omega x} \\ \psi_{III} = C e^{-\omega x} \end{array} \right)$$

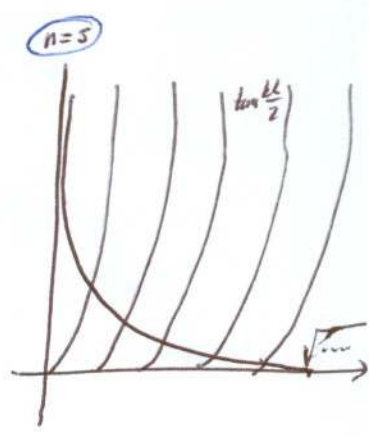
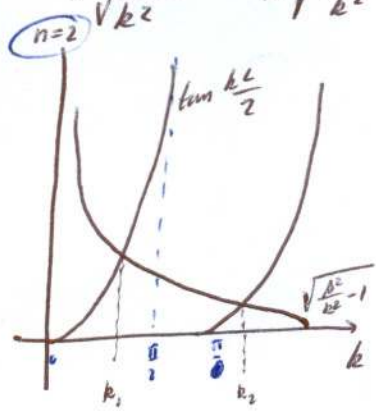
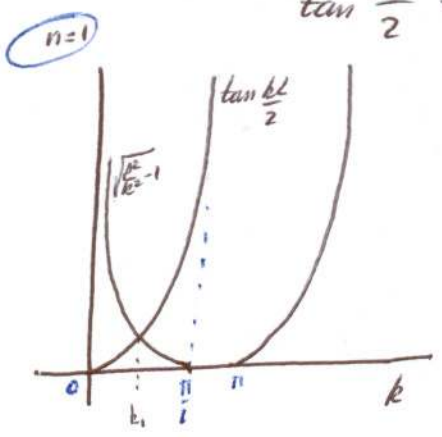
We can consider one boundary only because of the symmetry:

$$\psi_{II}(\frac{L}{2}) = \psi_{III}(x = \frac{L}{2}) \quad \& \quad \left. \frac{d\psi_{II}}{dx} \right|_{x=\frac{L}{2}} = \left. \frac{d\psi_{III}}{dx} \right|_{x=\frac{L}{2}}$$

$$\Leftrightarrow B \cos \frac{kL}{2} = C e^{-\omega \frac{L}{2}}, \quad -Bk \sin \frac{kL}{2} = -C\omega e^{-\omega \frac{L}{2}}$$

$$\tan \frac{kL}{2} = \frac{\omega}{k} = \frac{\sqrt{\beta^2 - k^2}}{\sqrt{k^2}} = \sqrt{\frac{\beta^2}{k^2} - 1}$$

intersect:



The number of intersections & thus solutions increase as β grows, i.e. as V_0 grows. This is consistent with the observations:

- $V_0 \rightarrow 0$ no ground state ~~no~~ solution
- $V_0 \rightarrow \infty$ an infinite number of sets: infinitely hard walls

We notice that for large enough values of $V_0 \ll \infty$, & small n , since the intersection goes higher as V_0 grows, ~~then~~ and $\tan \frac{kL}{2}$ is asymptotic to $\frac{kL}{2} \approx n \frac{\pi}{2}$ for large values

$$\Rightarrow \sqrt{\frac{\beta^2}{\hbar^2}} k^2 = \beta^2 - \omega^2 = \frac{2m}{\hbar^2} V_0$$

$$\left(\frac{k_n L}{2}\right)^2 \approx \left(\frac{n \pi}{2}\right)^2 \quad n \in 2\mathbb{Z}+1$$

$$\Leftrightarrow E_n \approx \frac{n^2 \pi^2 \hbar^2}{2m L^2} \quad \text{as for an infinitely hard-wall potential but only for } n \in 2\mathbb{Z}+1$$

- Odd solutions

$$\psi(x) = -\psi(-x)$$

$$\Rightarrow \psi_{II}(x) = A \sin kx \quad \& \quad C = F \quad \begin{pmatrix} \psi_I = C e^{\omega x} \\ \psi_{III} = -C e^{-\omega x} \end{pmatrix}$$

$$\text{boundary.c.} \Rightarrow A \sin \frac{kL}{2} = -C e^{-\omega \frac{L}{2}} \quad \& \quad A k \cos \frac{kL}{2} = C \omega e^{-\omega \frac{L}{2}}$$

$$\cot \frac{kL}{2} = -\frac{\omega}{k} \Leftrightarrow \tan\left(\frac{kL}{2} + \frac{\pi}{2}\right) = \sqrt{\frac{\beta^2}{k^2} - 1}$$

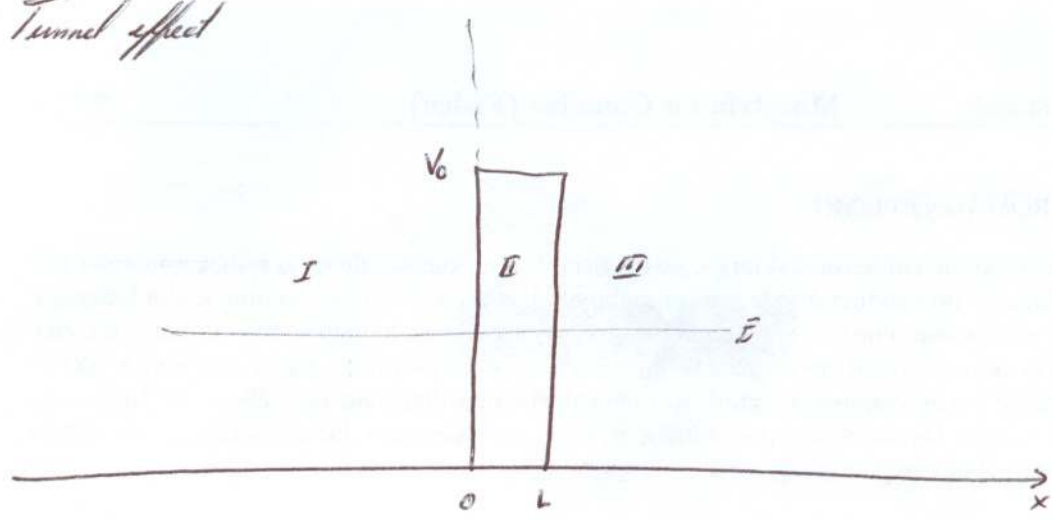
As before, for β large and n small

$$\left(\frac{kL}{2} + \frac{\pi}{2}\right)^2 \approx \left(\frac{(2n-1)\pi}{2}\right)^2 \quad n \in \mathbb{Z}$$

$$\left(\frac{kL}{2}\right)^2 \approx \pi^2 \left(n - \frac{1}{2}\right)^2$$

$$\frac{2m E_n}{\hbar^2} \frac{L^2}{4} = \pi^2 \left(n - \frac{1}{2}\right)^2 \Leftrightarrow E_n \approx \frac{\hbar^2 \left[\frac{(2n-1)\pi}{2}\right]^2}{2m L^2}$$

~ Tunnel effect



$$V = \begin{cases} V_0 & 0 \leq x \leq L \\ 0 & \text{otherwise} \end{cases}$$

Because we know the behavior of wavefunctions in regions, such as those presented by the potential, we can describe the behavior of particles incoming from $x \rightarrow -\infty$ with $E < V_0$, colliding the potential barrier and coming out, if possible, through region II until region III. It's clear that there is a reflected wavefunction in I, a transmitted one in region II, which shall decay exponentially & an emerging front in region III.

Under these circumstances, the wavefunctions should be

$$\begin{aligned} \psi_I &= \cancel{Ae^{ikx}} \underbrace{Ae^{-ikx}}_{\text{reflected}} + \underbrace{Be^{ikx}}_{\text{incoming}} & k^2 &= \frac{2mE}{\hbar^2} \\ \psi_{II} &= \underbrace{Ce^{-\omega x}}_{\text{in transmitted}} + \underbrace{De^{\omega x}}_{\text{bouncing in barrier}} & \omega^2 &= \frac{2m}{\hbar^2}(V_0 - E) \\ \psi_{III} &= \cancel{Ee^{-ikx}} + \underbrace{Fe^{ikx}}_{\text{transmitted}} \end{aligned}$$

In our case, ~~B~~ $E=0$ because we don't expect any incoming particles from the right. D is NOT zero because it can also bounce within the barrier

Applying the boundary conditions, we find

$$\psi_I(x=0) = \psi_{II}(x=0)$$

$$\psi_{II}(x=L) = \psi_{III}(x=L)$$

$$\left. \frac{d\psi_I}{dx} \right|_{x=0} = \left. \frac{d\psi_{II}}{dx} \right|_{x=0}$$

$$\left. \frac{d\psi_{II}}{dx} \right|_{x=L} = \left. \frac{d\psi_{III}}{dx} \right|_{x=L}$$

$$\Rightarrow A+B = C+D$$

$$Ce^{-\omega L} + De^{\omega L} = Fe^{ikL}$$

$$-ikA + ikB = -\omega C + \omega D$$

$$-\omega Ce^{-\omega L} + \omega De^{\omega L} = ikFe^{ikL}$$

Since there are 4 eqs. for 5 unknowns, the best we can do is to express B in terms of F , incident transmitted

$$\frac{B}{F} = \frac{1}{4} \left(\frac{k}{\omega} + \frac{\omega}{k} \right) e^{ikL-\omega L} - \frac{i}{4} \left(\frac{k}{\omega} - \frac{\omega}{k} \right) e^{ikL+\omega L} + \frac{1}{2} e^{ikL-\omega L} + \frac{1}{2} e^{ikL+\omega L}$$

$$\text{If } V_0 \gg E \rightarrow \frac{k}{\omega} \ll 1 \quad \& \quad e^{-\omega L} \ll 1$$

$$\Rightarrow \frac{B}{F} \approx \frac{i}{4} \frac{\omega}{k} e^{ikL+\omega L} + \frac{1}{2} e^{ikL+\omega L} \Rightarrow \frac{F}{B} = \frac{4e^{(-ik-\omega)L}}{2 + i\omega/k}$$

This expression is very convenient to measure the tunneling effects by computing the transmission probability of finding some particles of the incident beam in the transmitted wavefunction

$$P_t = \frac{|\psi_{trans}|^2}{|\psi_{inc}|^2} = \frac{|F|^2}{|B|^2} \approx \left| \frac{4e^{(-ik-\omega)L}}{2 + i\omega/k} \right|^2 = \frac{16e^{-2\omega L}}{4 + \frac{\omega^2}{k^2}}$$

The ~~more~~ exact result is:

$$P_t = \frac{16e^{-2\omega L}}{\frac{k^2}{\omega^2}(e^{2\omega L}-1)^2 + \frac{\omega^2}{k^2}(e^{2\omega L}-1)^2 + 2(1+6e^{2\omega L}+e^{4\omega L})}$$

- Clearly, as ω or L grows, P_t decreases very rapidly (exponentially).
- Also, if $k \rightarrow 0$, or $\omega \rightarrow \infty$ or $L \rightarrow \infty$, $P_t \rightarrow 0$

$\sim \alpha$ decay

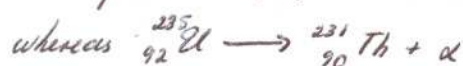
Heavy nuclei ($Z \geq 82$) decay as a means to improve their stability. Besides, it's a natural process, in contrast to emission of p's or n's, as the latter would require to invest some energy:

Energy released in a decay $Q_x = (m_i - m_f - \overset{\text{released-particle mass}}{m_x})c^2$

$Q_\alpha > 0$ whereas $Q_p, Q_n < 0$



$$\left. \begin{array}{l} m_{\text{U}} \sim 218.94 \text{ GeV}/c^2 \\ \text{Protactinium} \longrightarrow m_{\text{Pa}} \sim 218.01 \text{ GeV}/c^2 \\ m_p \sim 0.94 \text{ GeV}/c^2 \end{array} \right\} \Rightarrow Q \approx -10 \text{ MeV}/c^2 \quad \text{invest energy!}$$

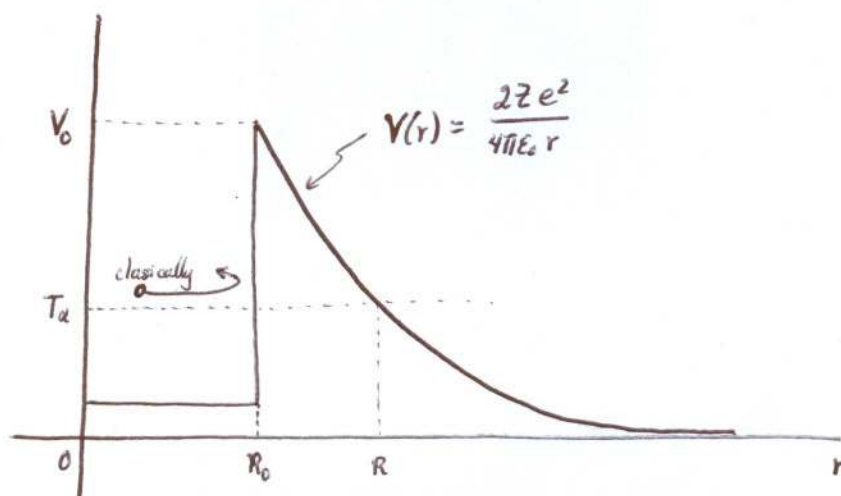


$$\left. \begin{array}{l} m_\alpha \sim 3.727 \text{ GeV}/c^2 \\ m_{\text{Th}} \sim 215.209 \text{ GeV}/c^2 \end{array} \right\} \Rightarrow Q \approx +4 \text{ MeV}/c^2 \quad \text{gain energy!}$$

Almost the whole Q is transferred to the kinetic energy of α

$$T_\alpha = \frac{A-4}{A} Q \quad (A: \text{mass number of initial nucleus})$$

The best model describing α -decay is by tunneling in the following potential:



- Priors:
- i) α -particle exists independently in the nucleus
 - ii) a potential barrier (strong force) keeps it in the nucleus
 - iii) there's a finite probability of tunneling P_t

We could estimate semi-classically the probability of tunneling by considering it to be ~~proportional~~ due to the bouncing on the "walls" of the nuclei. The probability of decay per second λ should be

$$\lambda = \nu P_t$$

$$\nu: \text{bouncing frequency of } \alpha\text{-particle} \rightarrow \nu = \frac{v_\alpha}{2R_0} = \frac{1}{R_0} \sqrt{\frac{T_\alpha}{2m_\alpha}}$$

$$\text{typically } v_\alpha \sim 2 \times 10^7 \text{ m/s } (T_\alpha \sim 8 \text{ MeV}), R_0 \sim 10^{-14} \text{ m} \Rightarrow \nu \sim 10^{21} \text{ Hz}$$

$$\text{Since in some cases, } t_{1/2}^{\text{half-life}} \sim 10^{10} \text{ y} \quad \& \quad t_{1/2} \propto \lambda^{-1} \quad \left(\text{in fact } t_{1/2} = \frac{\ln 2}{\lambda} \right)$$

$$\Rightarrow \underline{\underline{P_t \ll 1}}$$

$$\begin{aligned} N(t) &= N_0 e^{-\lambda t} \\ \frac{1}{2} N_0 &\equiv N_0 e^{-\lambda t_{1/2}} \end{aligned}$$

With the quantum formalism of tunneling, we can compute P_t . We found

$$P_t \sim e^{-2\omega L} \quad \omega^2 = \frac{2m}{\hbar^2} (V_0 - E)$$

Generalizing, we may approximate in our case the probability of tunneling up to r

$$\ln P_t(r) \sim -2 \int_{R_0}^r dr' \omega(r') \quad \text{with} \quad \omega^2(r) = \frac{2m_\alpha}{\hbar^2} (V(r) - T_\alpha)$$

Taking R as ~~fixed~~, the kinetic energy must satisfy

$$T_\alpha = V(R) = \frac{2Ze^2}{4\pi\epsilon_0 R} \Leftrightarrow R = \frac{2Ze^2}{4\pi\epsilon_0 T_\alpha}$$

$$\Rightarrow \omega(r) = \sqrt{\frac{2m}{\hbar^2}} \sqrt{\left(\frac{2Ze^2}{4\pi\epsilon_0} \right) \left(\frac{1}{r} - \frac{1}{R} \right)} = \sqrt{\frac{2m_\alpha T_\alpha}{\hbar^2}} \left(\frac{R}{r} - 1 \right)^{1/2}$$

Thus, the probability P_t is

$$\ln P_t \sim -2 \int_{R_0}^R dr \sqrt{\frac{2m_\alpha T_\alpha}{\hbar^2}} \left(\frac{R}{r} - 1 \right)^{1/2} = -2 \sqrt{\frac{2m_\alpha T_\alpha}{\hbar^2}} R \left[\cos^{-1} \sqrt{\frac{R_0}{R}} - \sqrt{\frac{R_0}{R}} \sqrt{1 - \frac{R_0}{R}} \right]$$

Supposing a broad band (before: $L \gg 1$)

$$R \gg R_0 \Rightarrow \cos^{-1} \sqrt{\frac{R_0}{R}} \approx \frac{\pi}{2} - \sqrt{\frac{R_0}{R}} \quad \& \quad \sqrt{1 - \frac{R_0}{R}} \approx 1$$

$$\Rightarrow \ln P_i \approx -2 \sqrt{\frac{2m T_\alpha}{\hbar^2}} R \left[\frac{\pi}{2} - 2 \sqrt{\frac{R_0}{R}} \right]$$

For terms of T_α , we obtain

$$\ln P_i \approx \frac{-2 \sqrt{2m_\alpha T_\alpha} Z e^2}{\hbar \pi \epsilon_0 T_\alpha} \left(\frac{\pi}{2} - 2 \sqrt{\frac{2\pi \epsilon_0 T_\alpha R_0}{Z e^2}} \right)$$

$$= - \underbrace{\sqrt{\frac{m_\alpha}{2}} \frac{e^2}{\epsilon_0 \hbar}}_{\sim 3 \times 10^{-7}} Z \sqrt{T_\alpha} + \frac{4e}{\hbar} \underbrace{\sqrt{\frac{m_\alpha R_0}{\pi \epsilon_0}}}_{\sim 2 \times 10^7} \sqrt{R_0} Z$$

$$= - \underbrace{\sqrt{\frac{m_\alpha}{2}} \frac{e^2 10^7 Z}{\epsilon_0 \hbar \sqrt{T_\alpha}}}_{\equiv \alpha \sim 3} + \underbrace{\frac{4e}{\hbar} \sqrt{\frac{m_\alpha 10^7}{\pi \epsilon_0}} \sqrt{R_0} Z}_{\equiv \beta \sim 2}$$

$$T_\alpha' = \frac{T_\alpha}{10^{-14}}, \quad R_0' = \frac{R_0}{10^{-14}}$$

Computing $\lambda = \nu P_i$

$$\Rightarrow \ln \lambda = \ln \nu + \ln P_i \approx \ln \left(\frac{\nu_\alpha}{2 R_0} \right) + \alpha \frac{Z}{\sqrt{T_\alpha}} + \beta \sqrt{R_0} Z$$

$\Rightarrow$ small differences in T_α cause large changes in λ

explaining why

$$t_{1/2} (^{232}\text{Th}) \sim 1.3 \times 10^{10} \text{ y}$$

$$T_\alpha \sim 4 \text{ MeV}$$

$$t_{1/2} (^{212}\text{Po}) \sim 3 \times 10^{-7} \text{ s}$$

$$T_\alpha \sim 9 \text{ MeV}$$

It fits well the data:

